

Performance Outcomes Following Isoinertial or Machine-Based Training Interventions in Female Junior Tennis Players

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Purpose: To evaluate changes in physical performance indicators after executing a flywheel or machine-based neuromuscular training program in young female tennis players. **Methods:** Twenty-four players were divided into a machine-based group (MG), flywheel group (FG), and control group. Countermovement jump (CMJ), sprint time (5, 10, and 15 m), change-of-direction (COD) performance (right and left), medicine-ball throws (overhead, forehand, and backhand), and serve velocity were included as tests at baseline, week 4, and week 8. **Results:** Both MG and FG largely improved from baseline to weeks 4 and 8 of training in CMJ (11.6, 10.6%; effect size [ES]=1.24, 1.95). Also, sprint time 5 m and 10 m, COD performance-right, and COD performance-left improved moderately to very largely at week 4 in MG (−5.0% to −6.2%; ES = −1.79 to −4.19) and FG (−2.9% to −5.1%; ES = −1.13 to −1.64), respectively. Regarding medicine-ball throw, only FG improved moderately to very largely from weeks 4 to 8 in overhead (9.3%; ES = 1.46), forehand (8.0%; ES = 1.08), and backhand (6.1%; ES = 1.15). Serve velocity improved moderately from weeks 4 to 8 in MG (5.8%; ES = 0.87). **Conclusions:** Four weeks of the tested programs seem sufficient to improve physical determinants in young female participants. Greater improvements in CMJ and medicine-ball throw following flywheel neuromuscular training indicate the importance of including exercises that emphasize the stretch-shortening cycle and involve the entire kinetic chain. Performing the same intervention with no load adjustments may stall or even decrease performance in some parameters from weeks 4 to 8.

Keywords: eccentric training, racket sports, neuromuscular performance, physical test

Women's tennis presents longer rallies alongside lower stroke and movement speeds compared with male competition.¹ Despite this, aspects such as intensity of play, distance covered, and overall number of strokes indicate that it remains a sport characterized by intense short bouts of exercise played between lower-intensity and longer resting periods.² It is generally accepted that men's tennis has evolved from a rather technical sport to a discipline determined by fast and powerful actions that can affect the outcome of the game.³ In women's tennis, the sport has suffered a similar evolution, and, for example, greater serve velocities can be observed in professional players and are increasing throughout competition.⁴ Furthermore, junior players with greater body mass, upper-body strength, and power will serve faster than those with lower fitness level indicators.^{4,5} In addition, sprints over short distances seem directly related to level and are especially determinant in junior female competitors.⁶ Thus, aspects such as speed, change of direction (COD), and power are important to compete effectively,⁵ are related to level,⁶ and have acquired growing importance among female players.

All these actions use the stretch-shortening cycle to increase velocity during execution.⁷ As these actions imply the efficient usage of prestretch and eccentric contraction of the muscle to use in the subsequent concentric phase, it would seem logical to aim training to improve these aspects. In this line, eccentrically

overloaded exercises have proven to evoke physiological responses⁸ that may lead to adaptations in the form of increases in overall strength, muscle power, hypertrophy, jumping ability, changes of direction, and speed.⁹ Particularly, devices that enhance the eccentric phase of contraction while performing resistance training, such as flywheels and rotational or conical pulleys, have been considered effective to improve high-intensity power-based actions.¹⁰ Based on several meta-analyses,^{11,12} a greater effectiveness of flywheel neuromuscular training (FNMT) over machine-based neuromuscular training (MNMT) (eg free weights and weight stack training) remains unclear. The need of further investigations is underlined as benefits on enhancing muscle strength, muscle structure, and increased performance adaptations of one method over another have not previously been documented.¹² Unfortunately, literature concerning the benefits of this type of training over more traditional methodologies and how responses may appear in adolescents is scarce,⁹ especially in women.¹³ In this regard, some investigations suggest a similar benefit as seen in men in power, strength, and muscle mass¹⁴ after following eccentric overloaded programs. Also, fatigue-related acute effects, such as soreness, strength loss, and muscle protein concentration in plasma, seem to similarly appear after comparable loads across sexes.¹⁵ Despite the fact that FNMT is an effective countermeasure for injury prevention and a potent incentive for physical enhancement in young and older female populations, a lack of clarity still exists with females related with dosages, intensity, and frequency factors of the prescriptive training.¹³ Contrary to other sports,^{16–18} no previous investigations have focused on analyzing the performance outcomes of following both training designs for 8 weeks in young female tennis players.

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Consequently, comparative interventions exploring the effectiveness of these approaches to improve physical determinants of tennis performance remain unstudied, especially in female populations. For this reason, the goal of this study was to evaluate changes in physical performance indicators after performing 8 weeks of either a flywheel or machine-based strength training program in young female tennis players. Both methods were performed alongside tennis-specific exercises (SE) at the end of the session. These included sprint, jump, and medicine ball throw (MBT) exercises (ie, combined training method) at maximum intensity. Based on previous literature,¹⁸ we hypothesized that the flywheel-based program would improve more thoroughly those actions that rely highly on the stretch-shortening cycle toward performance (ie, countermovement jump [CMJ]). Nevertheless, both groups would generally achieve similar improvements in 8 weeks of training.

Methods

Subjects

Twenty-four competitive female tennis players participated in this study (Table 1). This sample size was justified by a priori power analysis (using GPower version 3.1.9.5, University of Dusseldorf, Dusseldorf, Germany) for repeated-measures, within-between interaction for analysis of variance (ANOVA) introducing the following parameters: effect size (ES) index (0.31) assuming a large partial eta-squared (0.09), α error probability (.05), power (0.8), number of groups (3), and measurements (3), which resulted in a sample size of 24 subjects. There were no significant differences in the biometric and training characteristics between groups before the training intervention. Inclusion criteria included all players having at least 7 years of experience in tennis competition and at least 3 years following a structured resistance training program based on a wide variety of MNMT, body weights, and plyometric and accentuated eccentric loading exercises that sought to improve hypertrophy, strength, and power output. Subjects were previously familiarized with, well trained in, and used these eccentric exercises as a main part of their routines. Until the familiarization period, most of them had never used FNMT before. They performed it 3 times a week at their respective academies. Players did not perform any fitness sessions 2 weeks prior to the intervention. Exclusion criteria included having significant injuries or pain at the time of the intervention or having undergone surgery in the past 12 months. Along the study, players were not allowed to modify characteristics on the rackets (ie, string tension, hand width and grip strength, technique of their strokes) they used. Before the investigation, all the participants and their guardians were informed about the procedures, and they all provided and signed written informed consent. They were free to withdraw from this study without penalty at any time.

The investigation was performed following current ethical standards, established in the Declaration of Helsinki, and was granted approval by a relevant institutional ethics committee (45/2018).

Design

Two tennis clubs from same city with similar training methodology were used to form the groups. Participants were divided into 3 groups: control (CG; $n = 9$), machine-based training (MG; $n = 7$), and flywheel training (FG; $n = 8$). Players were randomly allocated into both experimental groups using stratified block randomization, but CG was conformed using nonrandomization. Pretest data indicated significant differences between MG and FG in sprint 10 and 15 m ($P < .05$) and between CG and FG in 5-0-5 right ($P < .01$) and left ($P < .05$) and serve velocity (SV) right and left ($P < .001$). CG was organized suitable to the club's needs, and this could explain differences. Two sessions based on the tests and the main part of the program were performed as familiarization sessions twice a week for 4 weeks prior the study. After these sessions and before the intervention, participants performed the pretests. In addition to the experimental training sessions, all participants took part in a 30-minute injury prevention program, including core exercises, shoulder rotator strength, mobility, and flexibility. These sessions were performed during the 8-week period, twice a week. Also, all players participated in a 2-hour long tennis training session 3 times a week. Training interventions performed alongside SE were performed on Mondays and Wednesdays after technical and tactical training. There were no dropouts, and no players sustained an injury or reported pain or discomfort as a result of the training program.

Methodology

The order of the tests was randomized to isolate the possible influence of fatigue. For data analysis, only the best result of attempts was selected. Before the start of the tests, players performed a standardized warm-up (ie, 15 min including general mobilization, specific and multidirectional tennis movements, and progressive speed-increasing exercises). A specific warm-up was executed before SV and MBT tests (ie, 8 progressive serves and different throws).

Jump Height. Participants performed a CMJ on a contact platform (Chronojump-Boscosystem) starting in a standing position with their hands on their hips and without arm swing, followed by a downward movement using a self-selected depth; participants then jumped as high as possible. Three attempts were executed with 1 minute of passive rest between them.

Sprint Time. Running sprint time was assessed for 5, 10, and 15 m using timing gates (Artek PNP response time < 0.5 ms at 1 kHz). The

Table 1 Participant Characteristics

	CG ($n = 9$)	MG ($n = 7$)	FG ($n = 8$)	Difference (P)
Age, y	15.6 (0.9)	15.6 (1.0)	15.8 (0.7)	.880
Height, cm	165.7 (6.7)	163.9 (8.5)	164.3 (6.3)	.704
Body mass, kg	60.5 (7.8)	54.9 (7.6)	52.7 (5.0)	.080
Training volume, h-wk ⁻¹	8.8 (0.3)	9.0 (0.0)	9.4 (1.1)	.174
Training background, y	8.7 (1.2)	9.7 (1.1)	9.9 (1.1)	.089

Abbreviations: CG, control group; FG, flywheel training group; MG, machine-based group. Note: Values are mean (SD), and P is value of the differences.

starting position was located 50 cm behind the starting line, and players were instructed to use a preferred starting position. Three attempts were executed with 2 minutes of passive rest between them.

Change-of-Direction Speed. The ability to perform a single and 180° COD over a 5-m distance by using a modified 5-0-5 right and left test was executed with maximal effort. Execution time was recorded with timing gates (Artek PNP response time <0.5 ms at 1 kHz). Two trials were performed on each side with 2 minutes of passive rest between them.

Serve Velocity. The SV was assessed on both sides of the court using a radar gun (Stalker Professional Sports Radar) and new tennis balls (Head ATP). The gun was calibrated and positioned on the center of the baseline, 3 to 4 m behind the player, and at the height of ball contact, following the fabricator's specifications. Five attempts at maximum speed with 25 seconds of rest between serves were required.

Medicine-Ball Throws. The 3-kg medicine ball maximal throwing speed on the overhead, forehand, and backhand movements was used. Overhead throws were performed from a line where players stood with both feet parallel side by side, somewhat apart, and holding the medicine ball behind the head. At that point, the ball was thrown forward as far as possible, hyperextending the back and shoulders without letting the player cross the line during and after the throw. Torso and hip rotation were also not allowed. In addition, forehand and backhand throws were executed standing in a closed-stance manner to the starting line. According to previous methods,⁵ they simulated a forehand–backhand stroke but throwing vigorously a medicine ball. Players were allowed to flex their legs, and torso and hip rotation was allowed, but they could not lose contact with the ground during and after the execution. A radar gun located 1 m behind the participant was used to determinate the speed of each attempt. Two attempts for every MBT were performed with 30 seconds of rest between them, and the fastest throw was used for data analysis. The research leader controlled the correct throwing techniques.

The reliability data of tests conducted are shown in Table 2. Sprint time, jump height, and SV showed excellent test–retest reliability (intraclass correlation coefficient [ICC] $\geq .90$; coefficient of variation $\leq 2.5\%$), whereas change of direction speed (CODS) and MBT showed high reliability (ICC $\geq .70$; coefficient of variation $\leq 4.5\%$). The standard error of measurement (SEM) and minimal detectable change were considered acceptable for all measurements.

Training Intervention. During the intervention program, CG only performed the mentioned tennis and injury prevention sessions for 8 weeks. In addition to that routine, MG and FG executed a 1-hour long FNMT (Figure 1A) or MNMT (Figure 1B), with 10 to 15 minutes of break in between sessions, 2 days per week, following loads expressed in Table 3A and B, respectively. Both interventions were followed by SE. FNMT program was performed using a rotational inertia device (Eccotek Training Force, Byomedic System SCP), consisting of a metal disk (diameter 0.42 m) with up to 16 masses (0.421 kg and 0.057 m diameter each one). The total moments of inertia ($I = mr^2$, where I = inertial, m = mass, and r = perpendicular distance from mass to the rotation center) for the rotational inertia device used to perform the exercises were $0.03 \text{ kg}\cdot\text{m}^2$ (from baseline to week 4) and $0.06 \text{ kg}\cdot\text{m}^2$ (from week 5 to week 8) based on a previous study.¹⁹ Following training sessions, each participant was asked to complete a self-assessment as an indicator of training intensity (ie, how players perceived their individual) using modified Borg rate of perceived exertion (Table 3).²⁰

Statistical Analysis

The values presented are expressed as mean (SD). The normality of the distributions and homogeneity of variances were assessed with the Shapiro–Wilk test. A 1-way ANOVA was used to assess significant differences in age, height, body mass, training volume and training background, and baseline test measurement among the 3 groups. To assess relative reliability, ICCs were calculated with the corresponding 95% confidence intervals. The ICC values were interpreted as follows: excellent ($>.90$), high ($.70$ – $.90$), moderate ($.50$ – $.69$), and low ($<.50$). Absolute reliability was calculated using the SEM.²¹ The SEM was used for calculating the minimal detectable change and was calculated as $\text{SEM} \times 1.96 \times \sqrt{2}$.²² Independent variables were defined in terms of the different strength training interventions (control vs machine-based vs flywheel) and the 3 measurement time points (pretest vs intertest vs posttest). The dependent variables were jump height (CMJ), sprint time (5, 10, and 15 m), CODS (5-0-5 right and left), SV (right and left), and MBT (forehand and backhand). In 2 way 3×3 repeated-measures ANOVA (3 times: pre–inter–posttest; 3 training groups: CG, MG, and FG) to test for global differences between pre–mid and postintervention, the 3 training groups and the interaction effect between 2 factors was applied. When a significant difference

Table 2 Reliability of Test Measurements

	CV, %	ICC (95% CI)	SEM	MDC
CMJ, cm	1.1	.995 (.980–.999)	0.30	0.83
Sprint 5 m, s	2.2	.918 (.693–.985)	0.03	0.08
Sprint 10 m, s	1.8	.929 (.734–.987)	0.04	0.11
Sprint 15 m, s	1.6	.944 (.793–.990)	0.05	0.14
5-0-5 right, s	1.8	.814 (.305–.965)	0.06	0.17
5-0-5 left, s	1.1	.805 (.273–.964)	0.04	0.11
SV right, $\text{km}\cdot\text{h}^{-1}$	1.6	.979 (.923–.996)	1.99	5.52
SV left, $\text{km}\cdot\text{h}^{-1}$	1.5	.992 (.992–.969)	1.75	4.85
Overhead MBT, $\text{km}\cdot\text{h}^{-1}$	3.0	.833 (.479–.959)	0.72	2.00
Forehand MBT, $\text{km}\cdot\text{h}^{-1}$	3.3	.773 (.292–.944)	0.79	2.19
Backhand MBT, $\text{km}\cdot\text{h}^{-1}$	2.9	.788 (.339–.9489)	0.64	1.77

Abbreviations: CMJ, countermovement jump; CV, coefficient of variation; ICC, intraclass correlation coefficient; MBT, medicine-ball throw; MDC, minimal detectable change; SEM, standard error of measurement; SV, serve velocity.

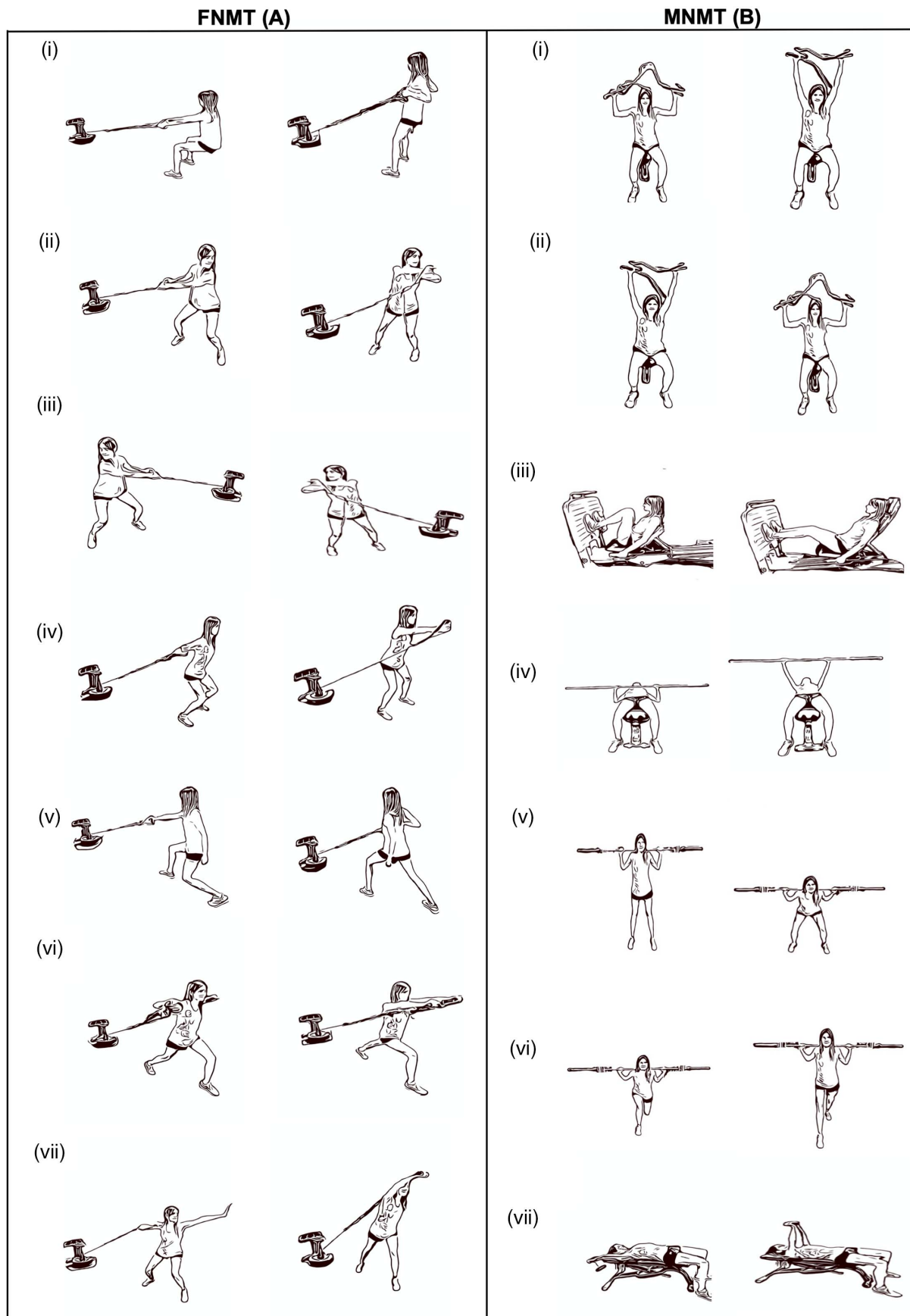


Figure 1 — The FNMT program intervention (A). (i) Low row 90°, (ii) forehand closed stance, (iii) backhand closed stance, (iv) 1-handed chest crossover, (v) 1-handed low row with 1 step, (vi) global chest press with 1 step, and (vii) 1-handed shoulder press. MNMT program intervention (B). (i) shoulder press machine, (ii) lateral pulldown machine, (iii) complete leg press machine, (iv) chest press, (v) half squat on multipower, (vi) forward lunge on multipower, and (vii) dumbbell pullover. FNMT indicates flywheel-based neuromuscular training; MNMT, machine-based neuromuscular training.

Table 3 Machine-Based (A) and Flywheel (B) Neuromuscular Training Intervention Programs

A. MNMT										
Wk	Sessions·wk ⁻¹ , n	Exercises, n	Sets, n	Reps, n	1RM, %	Intended velocity	Mean RPE	Rest set/round, min	Exercise program	SE
1–2	2	5	3	8	50–60	Max	5.4 ± 0.7	1.5/3	1. Bench press 2. Lateral pulldown machine	1. 2 × 5-, 10-, 15-m sprint 2. 2 × 5-m sprint go and return with left leg
3–4	2	5	3	8	50–60	Max	6.2 ± 0.5	1.5/3	3. Complete leg press machine 4. Shoulder press machine 5. Half squat on multipower with 2 legs	3. 2 × 5-m sprint go and return with right leg 4. 2 × 6 reps CMJ
5–6	2	5	3	6	60–70	Max	6.3 ± 0.6	1.5/3	1. Bench press 2. Lateral pulldown machine	1. 3 × 5-, 10-, 15-m sprint 2. 3 × 5-m sprint go and return with left leg
7–8	2	5	3	6	60–70	Max	6.2 ± 0.6	1.5/3	3. Complete leg press machine 4. Dumbbell pullover 5. Half squat on multipower with 1 leg	3. 3 × 5-m sprint go and return with right leg 4. 3 × 6 reps CMJ 5. 3 × 6 reps 2-hand throw with MB 3 kg (forehand, backhand, and overhead)
B. FNMT										
Wk	Sessions·wk ⁻¹ , n	Exercises, n	Sets, n	Reps, n	Overloads, n	Intended velocity concentric phase	Mean RPE	Rest set/round, min	Exercise program	SE
1–2	2	4	3	8	1	Max	5.6 ± 0.7	1.5/3	1. Dorsal 2 hands 90° flex	1. 2 × 5-, 10-, 15-m sprint
3–4	2	4	3	8	1	Max	6.2 ± 0.7	1.5/3	2. Forehand closed stance static 3. Backhand closed stance static 4. 1-hand chest crossover	2. 2 × 5-m sprint go and return with left leg 3. 2 × 5-m sprint go and return with right leg 4. 2 × 6 reps CMJ
5–6	2	5	3	6	2	Max	6.1 ± 0.7	1.5/3	1. 1-handed low row with 1 step	1. 3 × 5-, 10-, 15-m sprint
7–8	2	5	3	6	2	Max	6.3 ± 0.5	1.5/3	2. Global pectoral with 1 step 3. Forehand closed stance static 4. Backhand closed stance static 5. Serve simulation	2. 3 × 5-m sprint go and return with left leg 3. 3 × 5-m sprint go and return with right leg 4. 3 × 6 reps CMJ 5. 3 × 6 reps 2-hand throw with MB 3 kg (forehand, backhand, and overhead)

Abbreviations: CMJ, countermovement jump; FNMT, flywheel neuromuscular training; MB, medicine ball; MNMT, machine-based neuromuscular training; Reps, repetitions; RPE, rating of perceived exertion; SE, specific exercises; 1RM, 1-repetition maximum..

was found for either main effect (times or training group), a Bonferroni post hoc analysis was performed. Because MBT data were not normally distributed, the 3 conditions were compared using nonparametric analyses (Friedman followed by a Wilcoxon signed-rank test to determine the location of the significant differences). The partial eta-square (η^2) ES was calculated to evaluate the main and interaction effects of the ANOVA. The η^2 values of .01 to .05, .06 to .13, and $>.14$ indicate a small, medium, and large ES, respectively.²³ To determine the magnitude of within-group change in variables, a Cohen *d* ES was performed on pairwise comparisons. The criteria to interpret the magnitude of the ES were 0.0 to 0.2 trivial, 0.2 to 0.6 small, 0.6 to 1.2 moderate, 1.2 to 2.0 large, and >2.0 very large.²⁴ Statistical significance was accepted at an alpha level of $P \leq .05$. All statistical analyses were performed using IBM SPSS Statistics software (version 23.0, SPSS Inc).

Results

The summary of the performance measurements is plotted in Figures 2 and 3. Magnitude of changes at weeks 4 and 8 of intervention are expressed in Table 4.

Jump Height

A large significant training group by time interaction ($F=6.6$, $P<.001$, $\eta^2=.39$) was found in CMJ. The post hoc test showed significant increases in jump height in both FG and MG from baseline to week 4 (FG: 3.0 cm, 11.3%, $P<.001$, ES = 1.45; MG: 2.5 cm, 10.3%, $P=.001$, ES = 1.47) and from baseline to week 8 (FG: 2.8 cm, 10.6%, $P<.001$, ES = 1.95; MG: 2.9 cm, 11.6%, $P<.001$, ES = 1.24), whereas no changes from weeks 4 to 8 were observed in both groups. The CG did not show any change in this variable.

5-, 10-, and 15-m Sprint Time

A large significant training group by time interaction ($F=3.6$, $P=.013$, $\eta^2=.26$) was found in 10-m sprint time, and a large significant main effect of time was found in 5 m ($F=4.5$, $P=.017$, $\eta^2=.18$). Nonsignificant group by time interaction was found in 5 ($F=2.0$, $P=.106$, $\eta^2=.16$) and 15 m ($F=1.3$, $P=.274$, $\eta^2=.11$). The post hoc test showed significant sprint time improvements from baseline to week 4 in 5-m (MG: -0.08 s, -6.2% , $P<.001$, ES = -4.19 ; FG: -0.06 s, -5.1% , $P<.001$, ES = -1.64) and 10-m sprint times (MG: -0.10 s, -5.0% , $P<.001$, ES = -2.57 ; FG: -0.06 s, -2.9% , $P=.024$, ES = -1.14). Significant increases in the 10-m sprint time were found from weeks 4 to 8 in MG (0.06 s, 2.9%, $P=.011$, ES = 1.35) and FG (0.05 s, 2.4%, $P=.038$, ES = 0.95). CG did not show any change in this variable.

Change-of-Direction Speed

A large group by time interaction was found in 5-0-5 right ($F=4.8$, $P=.003$, $\eta^2=.31$), and a large significant main effect of time ($F=33.0$, $P<.001$, $\eta^2=.61$) was found for 5-0-5 left. Relating to 5-0-5 right, baseline to week 4 improvements were observed in response to MG (-0.17 s, -5.8% , $P<.001$, ES = -1.79) and FG (-0.09 s, -3.3% , $P=.009$, ES = -1.13), whereas from weeks 4 to 8, MG reduced their performance (0.12 s, 4.3%, $P=.001$, ES = 1.17), and no changes were found in FG (-0.05 s, 1.5%, $P=.119$, ES = 0.70). With respect to the 5-0-5 left, all 3 training groups significantly improved their performance from baseline to week 4 (MG: -0.17 s, -5.9% , $P<.001$, ES = -3.32 ; FG: -0.11 s, -3.9% ;

$P=.001$, ES = -1.63 ; CG: -0.09 s, -3.0% , $P=.005$, ES = -0.97). FG and CG also improved from baseline to week 8 (FG: -0.06 s, -2.2% , $P=.022$, ES = -1.72 ; CG: -0.06 s, -2.0% , $P=.021$, ES = -1.01), and MG reduced their performance from weeks 4 to 8 in 5-0-5 left (0.12 s, 4.5%, $P=.004$, ES = 1.01).

Serve Velocity

A large significant main effect of time was found for SV right ($F=5.5$, $P=.008$, $\eta^2=.21$) and left ($F=4.9$, $P=.012$, $\eta^2=.19$), whereas a nonsignificant group by time effect was found on both sides ($F=.32$, $P=.865$, $\eta^2=.03$ and $F=1.6$, $P=.200$, $\eta^2=.13$). The post hoc test showed significant increases in SV left from baseline to week 8 (5.9 km·h⁻¹, 4.6%, $P=.040$, ES = 0.79) and from weeks 4 to 8 (7.3 km·h⁻¹, 5.8%, $P=.016$, ES = 0.87) in the MG left serving side. Results indicated no changes in the right side of the MG or on both sides in the FG. CG did not show any change in this variable.

Medicine-Ball Throws

A Friedman ANOVA showed that there were significant differences between time conditions only in the FG in the overhead ($P=.002$), forehand ($P=.045$), and backhand ($P=.018$) MBT velocity. A series of Wilcoxon signed ranks tests showed that there were significant increases in overhead, forehand, and backhand MBT velocity from weeks 4 to 8 (2.1 km·h⁻¹, 9.3%, $P=.016$, ES = 1.46; 2.0 km·h⁻¹, 8.0%, $P=.034$, ES = 1.08; and 1.5 km·h⁻¹, 6.1%, $P=.024$, ES = 1.15) and from baseline to week 8 (2.8 km·h⁻¹, 12.4%, $P=.011$, ES = 3.10; 2.4 km·h⁻¹, 9.6%, $P=.028$, ES = 1.15; 2.4 km·h⁻¹, 10.1%, $P=.026$, ES = 1.24) in FG, whereas no differences were found from baseline to week 4. CG and MG did not show any change in this variable.

Discussion

This study aimed at evaluating changes in performance variables after performing 8 weeks of either an FNMT or MNMT program in young female tennis players. Results in this investigation indicate that 4 weeks of either methodology improved jumping height, sprint time, and CODS. Specifically, regarding group differences, the FNMT elicited slightly larger gains in CMJ (FG: 11.3%; ES: 1.45 vs MG: 10.3%; ES: 1.47) and larger improvements in all MBT (FG: 8.0%–12.4%; ES: 1.08–3.10 vs MG: 0.6%–9.3%; ES: 0.08–0.87) than the MNMT after 4 and 8 weeks of training, respectively. Conversely, MNMT was more effective than the FNMT model to improve 10-m sprint (MG: -5% ; ES: -2.57 vs FG: -2.9% ; ES: -1.14) and COD 5-0-5 right (MG: -5.8% ; ES: -1.79 vs FG: -3.3% ; ES: -1.13) performance within the first 4 weeks of training. MBTs were improved in posttest measurements only in the FG, whereas SV slightly improved in the MG left, although changes were comparable across groups.

Although both programs successfully developed jumping capacity in 4 weeks of training and maintained values up to 8 weeks, group by time interactions showed a greater improvement in jump height in the FNMT over MNMT (3.8 cm vs 2.9 cm, respectively). This is in accordance with previous investigations^{18,25} that found flywheel-based training induced superior improvements in power-related actions, such as the CMJ, whereas overall strength gains did not necessarily follow the same outcome.¹⁷ Physiologically, favoring the eccentric phase of contraction while performing the program could lead to specific neuromuscular adaptations, which include a greater recruitment

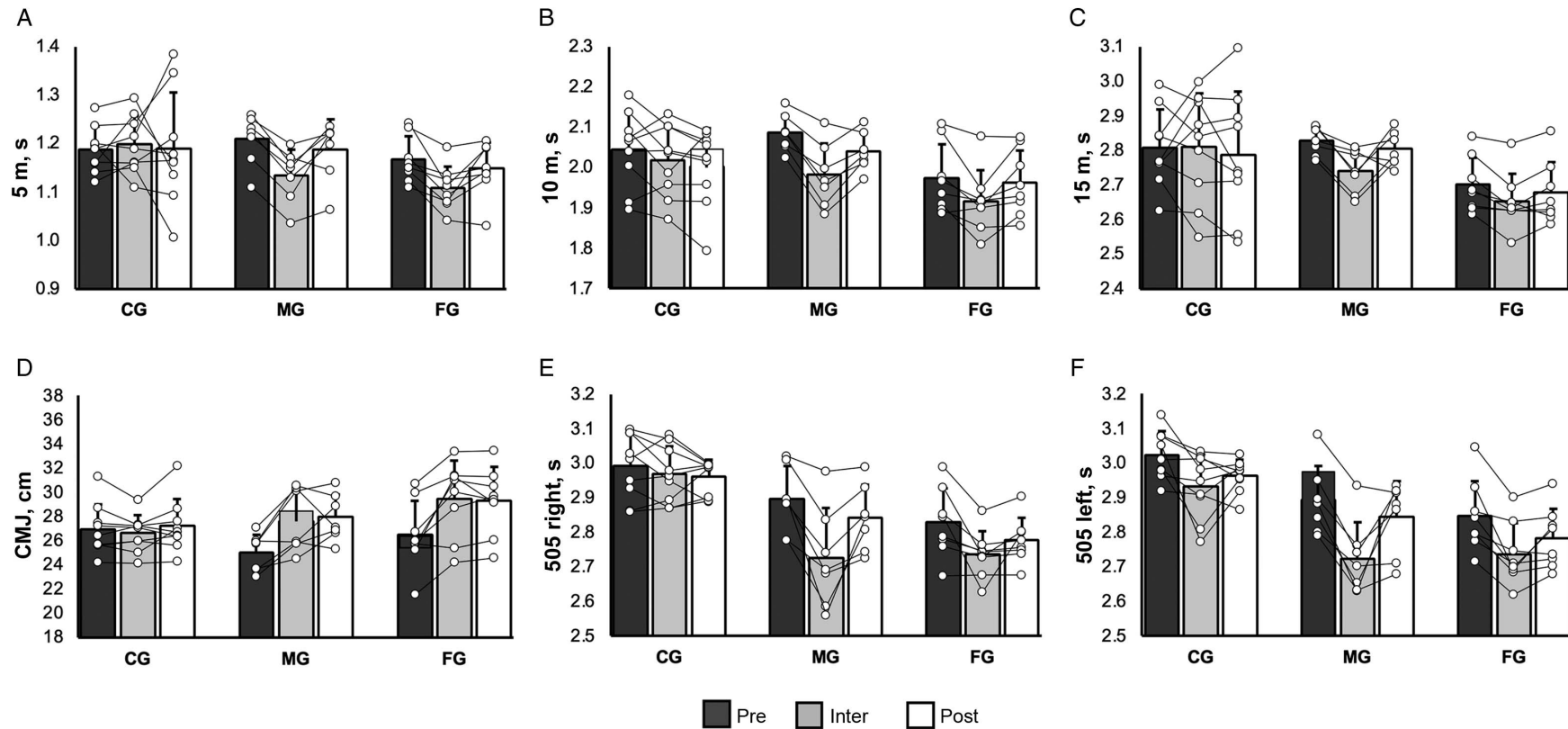


Figure 2 — Group average and individual points jump height, sprint time (5, 10, and 15 m), and change-of-direction performance (5-0-5 right and left) for MG, FG, and CG at baseline (pre), week 4 (inter), and week 8 (post) of the training intervention. (A) 5-m sprint time, (B) 10-m sprint time, (C) 15-m sprint time, (D) CMJ, (E) 5-0-5 right, and (F) 5-0-5 left. CG indicates control group; CMJ, countermovement jump; FG, flywheel training group; MG, machine-based training group.

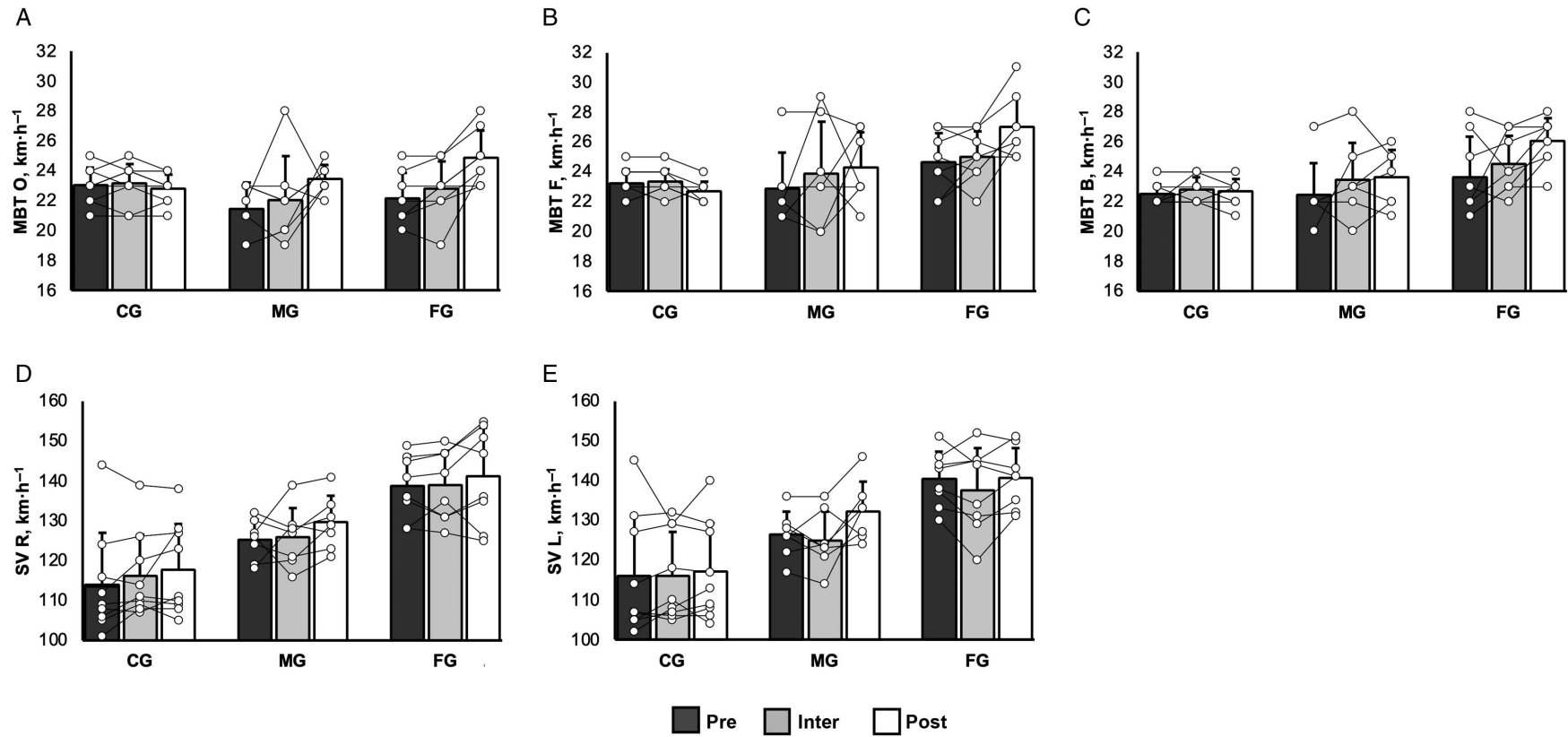


Figure 3 — Group average and individual points SV (SV right and left) and MBT (overhead, forehand, and backhand) for MG, FG, and CG at baseline (pre), week 4 (inter), and week 8 (post) of the training intervention. (A) MBT O; (B) MBT F; (C) MBT B; (D) SV R; (E) SV L. CG indicates control group; FG, flywheel group; MBT, medicine-ball throw; MBT B, MBT backhand; MBT F, MBT forehand; MBT O, MBT overhead; MG, machine-based group; SV, serve velocity; SV L, SV left; SV R, SV right.

Table 4 Changes in the Performance Variables Obtained During Pre, Inter, and Post Strength Training Intervention

	Control group (n = 9)			Machine-based group (n = 7)			Flywheel group (n = 8)		
	ES	Descriptor	Change, %	ES	Descriptor	Change, %	ES	Descriptor	Change, %
Jump height									
CMJ, cm									
Inter-pre	-0.35	Small	-1.2	1.47**	Large	10.3	1.45***	Large	11.3
Post-pre	0.36	Small	1.1	1.24***	Large	11.6	1.95***	Large	10.6
Post-inter	0.50	Small	2.2	-0.16	Trivial	1.2	-0.17	Trivial	-0.6
Sprint time									
Sprint 5 m, s									
Inter-pre	0.22	Small	0.9	-4.19***	Very large	-6.2	-1.64***	Large	-5.1
Post-pre	0.00	Trivial	0.0	-0.46	Small	-1.8	-0.40	Small	-1.7
Post-inter	-0.10	Trivial	-0.9	1.06	Moderate	4.7	1.00	Moderate	3.6
Sprint 10 m, s									
Inter-pre	-0.41	Small	-1.3	-2.57***	Very large	-5.0	-1.14*	Moderate	-2.9
Post-pre	0.71	Moderate	-2.2	-1.73	Large	-2.2	-0.21	Small	-0.6
Post-inter	0.34	Small	-0.8	1.35*	Large	2.9	0.95*	Moderate	2.4
Sprint 15 m, s									
Inter-pre	0.02	Trivial	0.1	-1.76	Large	-3.1	-1.36	Large	-1.8
Post-pre	-0.12	Trivial	-0.7	-0.75	Moderate	-0.9	-0.98	Moderate	-0.9
Post-inter	-0.26	Small	-0.8	0.79	Moderate	2.3	0.64	Moderate	0.9
Agility									
5-0-5 right, s									
Inter-pre	-0.39	Small	-0.7	-1.79***	Large	-5.8	-1.13**	Moderate	-3.3
Post-pre	-0.36	Small	-1.0	-0.84	Moderate	-1.8	-1.23	Large	-1.8
Post-inter	-0.14	Trivial	-0.3	1.17**	Moderate	4.3	0.70	Moderate	1.5
5-0-5 left, s									
Inter-pre	-0.97**	Moderate	-3.0	-3.32***	Very large	-5.9	-1.63**	Large	-3.9
Post-pre	-1.01*	Moderate	-2.0	-0.60	Moderate	-1.7	-1.72*	Large	-2.2
Post-inter	0.38	Small	1.0	1.01**	Moderate	4.5	0.94	Moderate	1.7
SV									
Right, km·h ⁻¹									
Inter-pre	0.55	Small	2.4	0.07	Trivial	0.5	0.07	Trivial	0.2
Post-pre	0.79	Moderate	3.8	0.62	Moderate	3.4	0.46	Small	1.9
Post-inter	0.32	Small	1.4	1.02	Moderate	3.0	0.37	Small	1.7
Left, km·h ⁻¹									
Inter-pre	0.02	Trivial	0.1	-0.30	Small	-1.1	-0.50	Small	-2.0
Post-pre	0.24	Small	1.0	0.79*	Moderate	4.6	0.05	Trivial	0.2
Post-inter	0.20	Small	0.9	0.87*	Moderate	5.8	0.58	Small	2.2
MBT									
Overhead, km·h ⁻¹									
Inter-pre	0.12	Trivial	0.5	0.22	Small	2.7	0.59	Small	2.8
Post-pre	-0.33	Small	-1.0	0.87	Moderate	9.3	3.10*	Very large	12.4
Post-inter	-0.47	Small	-1.4	0.42	Small	6.5	1.46*	Large	9.3
Forehand, km·h ⁻¹									
Inter-pre	0.19	Trivial	0.5	0.33	Small	4.4	0.22	Small	1.5
Post-pre	-0.55	Small	-2.4	0.64	Moderate	6.3	1.15*	Moderate	9.6
Post-inter	-0.77	Moderate	-2.9	0.10	Trivial	1.8	1.08*	Moderate	8.0
Backhand, km·h ⁻¹									
Inter-pre	0.39	Small	1.5	0.48	Small	4.5	0.35	Small	3.7
Post-pre	0.23	Small	1.0	0.42	Small	5.1	1.24*	Large	10.1
Post-inter	-0.14	Trivial	-0.5	0.08	Trivial	0.6	1.15*	Moderate	6.1

Abbreviations: CMJ, countermovement jump; ES, effect size; MBT, medicine-ball throw; SV, serve velocity. Note: Magnitudes of ESs were assessed using the following criteria: 0.0–0.2 = trivial, 0.2–0.6 = small, 0.6–1.2 = moderate, 1.2–2.0 = large, and >2.0 = very large.

* $P < .05$. ** $P < .01$. *** $P < .001$ significant difference within groups.

of higher threshold motor units and higher cortical excitability.²⁶ In this regard, the FNMT would benefit and take advantage of energy storage in the eccentric phase of contraction and lead to greater power outputs in the subsequent concentric phase.²⁷ Although the aforementioned studies focused on participants of different genders, it would seem logical to expect similar outcomes following a FNMT regarding jump height. Literature suggests that effects would be similar to those observed in men following accentuated eccentric programs as similar benefits and outcomes are observed regarding power, strength, and muscle mass.¹⁴

Increases in sprint performance following programs were observed in both groups performing 4 weeks of FNMT or MNMT, whereas some results returned to initial values (as in 10-m sprint time) at 8 weeks of intervention. Strength training, regardless the preferred methodology, has previously been considered effective to increase speed in young athletes^{10,16} and junior tennis players.^{18,28} Therefore, in accordance with previous studies, either an FNMT or MNMT program seemed to be an effective tool to enhance sprints over short distances in young female tennis players after 4 weeks of intervention. These results follow the general idea that nonspecific methods such as resistance training can improve sprinting due to enhanced strength and power production alongside changes in muscle architecture, especially affecting initial phases of acceleration.²⁹ In addition, the specific layout that participants followed in this investigation (ie, SE following strength tasks) could make the intervention be considered as a combined method. In this scenario, training could have offered greater effects as the benefits of strength training would add to those seen after performing specific sprint training, such as increased step length and reduced contact time.²⁹ Regarding group differences, although the 2 programs analyzed demonstrated improvements in 5 and 10 m at week 4, the MG (−5.0%; ES: −2.57) showed greater increases than the FG (−2.9%; ES: −1.14) in 10 m. This may have been induced by a slightly higher number of exercises targeting specifically the lower body in the MNMT over the FNMT (ie, leg press machine and half squat). Also, the differences in the initial values between the MG and FG groups (see Figure 2B) could indicate that MG was more sensitive to training-induced adaptations following the program. In relation to values returning to initial values from weeks 4 to 8, reasons could be the inadequate order in the planned training sessions and 3-hour long duration of each training session (tennis and intervention) conducted with a small break in between sessions of 10 to 15 minutes. Young players could be highly affected by load accumulations, enhanced metabolic stress, and fatigued state compromising physiological responses and adaptations,³⁰ especially regarding predominantly eccentric interventions. Other reasons could be given for these results; although intensity (inertial overloads) increased from weeks 1–4 to 5–8, and bilateral predominance shifted to unilateral exercises, these could be considered insufficient changes to observe further improvements following the second part of the intervention. Previously, investigations observed no changes in sprinting actions following periodized strength training during several months due to the need of greater variability, load adjustments, and increases in specific sprinting volume.³¹

Four weeks of either FNMT or MNMT programs seem sufficient to achieve significant improvements in CODS. Previous investigations have found the use of flywheels highly effective to increase CODS.^{16,18} Considering the specific demands of these actions, which typically involve a powerful and fast transition between eccentric and concentric phases, it would seem logical to expect greater improvements in the FG. Exercises present in the

FNMT performed using flywheel devices accentuate the effort in the braking phases of movement,¹⁶ possibly explaining adaptations seen in this group. Nevertheless, greater improvements were observed in the MG compared with FG from baseline to week 4. The greater number of exercises targeting the lower body in the MNMT program could explain these differences. Also, SE performed after the strength interventions in both groups could have offered adaptations on multiple determinant factors upon which CODS depends³² as the interventions could be considered as combined training methods and included specific sprinting and COD actions. Results from weeks 4 to 8 indicate maintenance of values in the FG, whereas MG returned to pretest values. As previously seen in male participants,¹⁸ applying an NMT for periods longer than 4 weeks without exercise variability and load adjustments and especially following technical and tactical sessions³⁰ may impair gains as young players could be highly affected by accumulations in load and an enhanced fatigued state.

The FG showed moderate to large improvements on all MBT at week 8 of intervention, whereas MG did not improve values in these variables. Previous investigations have shown the effectiveness of plyometric training and throwing-based programs on MBT performance and upper-body power.³³ Results here are in accordance with the mentioned studies, although certain differences regarding the program and participants can be stated. Interventions including exercises that emphasize the stretch-shortening cycle nature of tennis actions but do not include a certain volume of exercises that involve the entire kinetic chain and are performed at relatively high speeds may fail to elicit movement-specific adaptations.²⁸ In this line, improvements appeared only after MBT exercises were included at week 4 in the SE offered to players at the end of the session, restating the importance of including tasks that involve a transfer of forces from lower to upper body.³⁴ Between-group differences showing the FNMT to be more effective than MNMT may be explained by an emphasized eccentric phase of contraction and taking advantage of energy storage while performing the flywheel-based exercises. The FNMT, compared with MNMT, presented tasks with a greater summation of forces from lower-body segments to upper-body extremities, added to a favored eccentric phase, leading to greater power outputs in the concentric phase.²⁷ In a similar way to results regarding CMJ, FNMT could induce superior improvements in power-related actions, such as MBT, which, added to the stimulus offered by SE, could have induced greater gains.

Surprisingly, in functional measurements such as the SV, results indicate a significant increase in performance at weeks 4 (4.6%, ES = 0.79) and 8 (5.8%, ES = 0.87) in the MG left serving side. Results can seem unexpected as MG showed no improvements in a highly related test, such as the MBT, whereas the FG presented gains in all throws. The tennis serve is considered as an action of multifactorial nature, relying on different physical aspects, such as anthropometry, technique, biomechanics, strength, power, and range of movement.⁴ As previously seen in the literature, the capacity of a player to produce velocity to the serve relies on these aspects, but the contribution of each variable may vary between individuals, ages, and playing levels.³⁵ The fact that upper-body power values tested via MBT increased following a training intervention might not reflect gains in a complex motor action such as the serve as it relies on several other involved aspects. Nonetheless, it has been suggested that the contribution of strength and power values over technical and biomechanical proficiency seems to gain importance as the player increases in age and level.³⁵ In short, this suggests that gains observed in the MG may be due to improvements

achieved through different interactions between the determinant physical factors that affect SV rather than the programs tested here.

This study had some limitations. First, as we have stated in the study design, pretest data showed differences between groups in sprint 10 and 15 m, CODS, and SV. Considering that the main goal of the study is to evaluate changes in physical performance indicators after performing a flywheel or machine-based strength training program, these initial performance differences may exert some disparate training adaptations on each group. Second, the fact that both programs did not follow the exact same exercises with 2 different equipments may have affected results. Nevertheless, frequency, total number of exercises, training volume, and reported perceived effort indicate that loads between groups were of a comparable magnitude. In addition, the intention was to compare 2 practical program designs that coaches may follow when implementing strength and conditioning sessions. Also, maturity status was not measured, and biological age could have affected results to some extent. The analysis of similar interventions in different ages and levels would be of great interest. Finally, the order of sessions (after technical–tactical practice) may have affected results to some extent.

Practical Applications

High-performance tennis has limited training availability due to constant travel and competition. The results in this study highlight the effectiveness of the 2 tested NMT interventions to achieve gains in key physical factors affecting performance in short periods of 4 weeks. In general, similar effects can be observed when comparing both methodologies. Alongside this, the study of adaptations following flywheel-based interventions may bridge the gap toward establishing specific load prescriptions in the young tennis player. Nevertheless, eccentrically overloaded programs should be applied with caution and under professional supervision, ensuring that athletes have a strong weightlifting background and a well-developed maturational status.

Conclusions

In conclusion, 4 weeks of FNMT or MNMT improved jumping height, sprinting over short distances, and CODS in young female participants. Performing the same intervention for longer periods showed some performance decreases (ie, sprint 10 m in MG and FG and 5-0-5 left and right in MG) and stagnation (ie, CMJ and sprint 5 m in MG and FG, 5-0-5 left and right in FG) in both training interventions from week 4 to week 8. MBTs were improved in posttest measurements only in the FG, whereas SV only improved in the MG. The order of the training sessions alongside load and exercise variability seems to be an essential component to observe further improvements in longer programs of up to 8 weeks.

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